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EGOR GALKIN

QUANT RESEARCHER · AI RISK ENGINEER

Quant researcher at the intersection of regulatory risk management and applied ML. **3 years at the Bank of Russia**: validation of IRB credit-risk models, fair-value assessment of derivatives under IFRS 13, and valuation-process automation. In parallel, I fine-tune a domain-specific SLM (Phi-4-mini → **93.3% tool-use**) and maintain an open-source bilingual wiki of **400+ articles** on stochastic calculus, quantum field theory, and quantitative finance.

KEY ACHIEVEMENTS

- ★ **Fine-tuned Microsoft Phi-4-mini (3.8B)** to **93.3% tool-use accuracy** on Apple M5 Pro via MLX, +25.6 pp over baseline. Tool-Use SFT, RAFT, CoT distillation, KTO, LoRA Routing (3 adapters), C++ via pybind11.
- ★ **Validated IRB credit-risk models (PD/LGD/EAD)** of systemically important banks under Bank of Russia Regulation 483-P: backtesting, discriminatory power, population stability — identified methodological deviations affecting capital adequacy ratios.
- ★ **Automated fair-value assessment of derivatives** under IFRS 13 (Level 2/3) with Python + QuantLib: interest-rate swaps, barrier options, structured notes.
- ★ **Accelerated quantitative engines** (GARCH, HMM, Monte Carlo) by **200×+** with C++ extensions (pybind11) and parallel computation (OpenMP).
- ★ **LLM Wiki** — open-source bilingual knowledge base of **400+ articles** on quant finance, stochastic calculus, QFT, and LLMs. Obsidian vault → React SPA with 3D force-graph visualisation.

EXPERIENCE · 6 YEARS 5 MONTHS

Risk Analyst · Risk Analysis Service — MAY 2025 → PRESENT

Central Bank of the Russian Federation, Moscow

- Authored **fair-value opinions on securities and derivatives** (IFRS 13/9), used in supervisory decisions for credit and non-credit financial institutions.
- Built a **fair-value automation pipeline** in Python + QuantLib for the Level 2/3 portfolio: interest-rate swaps, barrier options, structured notes, FX forwards.
- Assessed the reliability of FV estimates for assets carrying market risk; ran **marketability checks** on financial-market operations and flagged price deviations from benchmarks.

Risk Manager · Supervision of Systemically Important Banks (IRB) — FEB 2024 → MAY 2025

Central Bank of the Russian Federation, Moscow

- Validated **credit-risk models** (PD/LGD/EAD) of systemically important banks under Bank of Russia Regulation 483-P — feeding capital adequacy calculations.
- Ran **statistical tests on rating systems**: PD backtesting, discriminatory power (Gini, AUC), population stability, calibration tests; econometric modelling on large credit-history datasets.
- Participated in **Bank of Russia Working Groups** on practical assessment of IRB models and in developing the methodology for evaluating IRB-model quality.

Risk Analyst · Risk Analysis Service — SEP 2023 → FEB 2024

Central Bank of the Russian Federation, Moscow

- Produced **fair-value opinions on financial instruments** (IFRS 13/9), including illiquid positions and instruments without an active market.
- Authored analytical opinions on the risk of securities holdings, with detailed analysis of issuer and counterparty financial position.
- Contributed to the **first cycle of FV-process automation** — laid the foundation for the pipeline subsequently rolled out across the Service.

Expert · Banking Supervision Service — DEC 2022 → JUL 2023

Central Bank of the Russian Federation, Moscow

- Performed **express analysis of credit-institution reporting** (RAS/IFRS): balance-sheet structure, P&L, asset and liability quality.
- Assessed **business models and growth strategies** of banks and non-bank credit institutions; identified the activities most exposed to risk.
- Applied Bank of Russia regulations (**590-P, 611-P, 199-I, 86-FZ**); evaluated risk-management quality, internal controls, and borrower investment projects.

STACK

ML / AI
PyTorch · MLX · Transformers · LoRA · QLoRA · RAFT · KTO · Tool-Use SFT · Claude · Phi-4

QUANT
QuantLib · Monte Carlo · GARCH · HMM · Itô calculus · Heston · Lévy · Copulas

LANGUAGES
Python · C++ · pybind11 · OpenMP · TypeScript · SQL

DATA / INFRA
Pandas · NumPy · SciPy · FastAPI · Docker · React · MOEX API · Bloomberg

REGULATORY
IFRS 13 · IFRS 9 · IRB · 483-P · 199-I · 590-P · 611-P · 86-FZ

EDUCATION

MSc · Stochastic Modelling in Economics & Finance
HSE University · Faculty of Economic Sciences
2024 → 2026

BSc · Mathematical Modelling in Economics
Bauman Moscow State Technical University
2019 → 2024

RESEARCH

Central-Bank Inflation Targeting
Monetary-policy reaction functions to inflation-expectation shocks — Taylor rules, DSGE estimation, regime robustness under structural breaks.

Non-Gaussian Models for Structured-Product FV
Heavy-tailed and jump-augmented models (SV, Lévy, jump-diffusion) vs the log-normal benchmark for path-dependent and barrier payoffs under IFRS 13.

MSc Thesis (HSE)
"Volatility Surface Construction as a Pricing Tool for Non-Linear Derivatives".

PROJECTS

Zeta Terminal
AI-native financial workspace: real-time data + QuantLib + LLM agents. Python, TypeScript, FastAPI, Claude, MLX.

Ultima Thule
Travel & exploration app: smart itineraries, offline maps, contextual place discovery. React Native, Supabase, Mapbox.

LLM Wiki (open-source)
400+ bilingual articles on quant finance, QFT, and LLMs. Obsidian vault → React SPA with 3D force-graph.

LANGUAGES

Russian	native
English	B2 · technical